# IAM-09: Troubleshooting Access Issues

> **Series:** IAM — IAM Administration | **Notebook:** 9 of 12 | **Created:** January 2026 | **Last Updated:** 10/12/2026

## Systematic Diagnosis of IAM Problems

Access issues are among the most common support requests. This notebook provides a systematic methodology for diagnosing and resolving IAM-related problems including permission denials, policy conflicts, and boundary issues.

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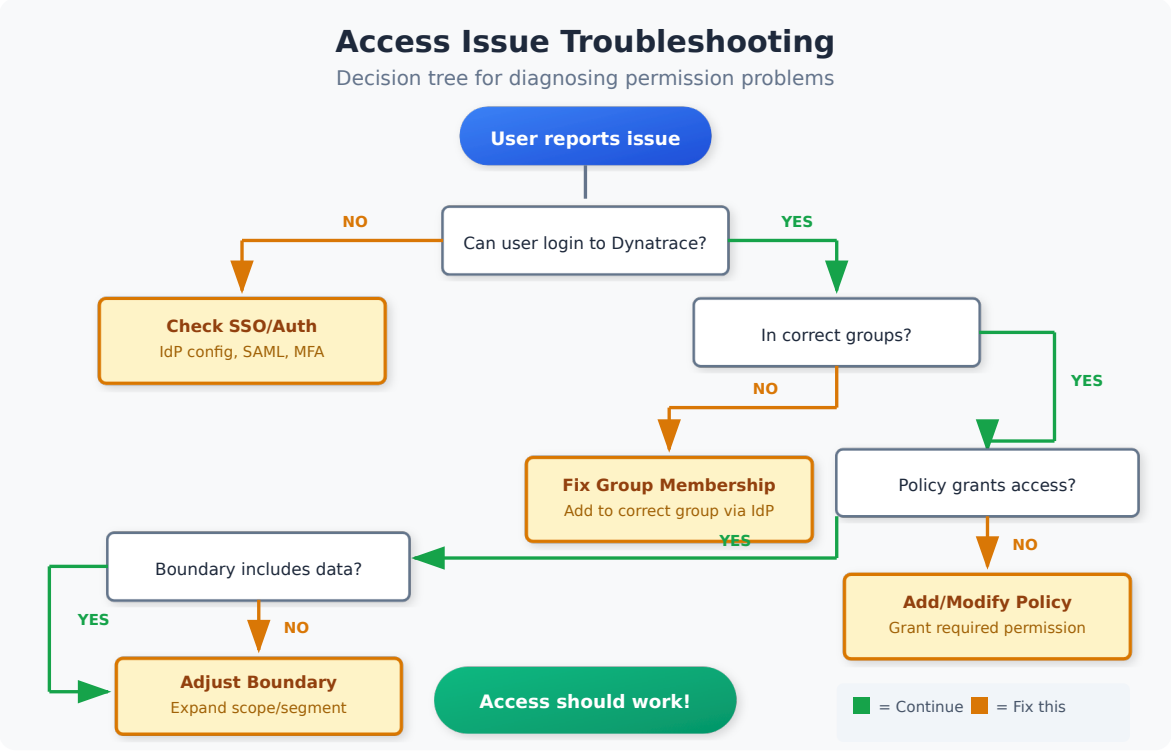

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# Prerequisites

| Requirement | Details |
|---|---|
| **Dynatrace Environment** | SaaS with Gen3 IAM |
| **Permissions** | `account-user-management` for full diagnosis |
| **Knowledge** | IAM-01 through 08 completed |

# 1. Troubleshooting Methodology

Follow a systematic approach to diagnose access issues.




## Step-by-Step Methodology

| Step | Check | Tools |
|---|---|---|
| 1 | User can authenticate | SSO logs, IdP logs |
| 2 | User exists in Dynatrace | Account Management → Users |

| Step | Check | Tools |
|---|---|---|
| 3 | User is in correct groups | Account Management → Groups |
| 4 | Groups have correct policies | Account Management → Policies |
| 5 | Policies grant required permissions | Policy viewer |
| 6 | Boundaries don't restrict access | Boundary configuration |
| 7 | No conflicting policies | Policy evaluation |
| 8 | Data exists and is accessible | DQL query |

### Information Gathering

Before troubleshooting, collect:

```
User Information:
☐ User's email address
☐ What action they're trying to perform
☐ Exact error message received
☐ When the issue started
☐ What changed recently (role, team, etc.)

Environment Information:
☐ Which environment(s) affected
☐ What data/resource they're accessing
☐ Screenshot of error (if available)
```

## 2. Common Access Issues

### Issue Categories

| Category | Symptoms | Common Cause |
|---|---|---|
| **Authentication** | Can't login | SSO misconfigured, account disabled |
| **Authorization** | Access denied | Missing policy, wrong group |
| **Data Access** | No results | Boundary restriction, no data |
| **Partial Access** | See some, not all | Boundary or policy condition |

| Category | Symptoms | Common Cause |
|---|---|---|
| **Intermittent** | Sometimes works | Caching, eventual consistency |

### Quick Diagnosis Table

| Symptom | Likely Cause | First Check |
|---|---|---|
| "Access Denied" | Missing permission | Policy assignments |
| "Forbidden" | Boundary restriction | Boundary config |
| "Not Found" | No data or no access | Query without filters |
| "Unauthorized" | Token/auth issue | Token validity |
| Can't see entity | Boundary or policy condition | Entity's `dt.security_context` vs. the boundary |
| Can read, can't write | Read-only policy | Policy permissions |

## 3. Permission Denied Debugging

When users receive "Access Denied" or "Forbidden" errors.

### Diagnostic Steps

**Step 1: Verify User Exists**

1. Go to Account Management → Users
2. Search for user by email
3. Verify account is active (not disabled)

**Step 2: Check Group Membership**

1. Click on user to view details
2. Note all groups user belongs to
3. Verify expected groups are present

**Step 3: Review Group Policies**

1. For each group, check assigned policies
2. Verify policies grant needed permissions
3. Check policy conditions/restrictions

**Step 4: Check Boundaries**

1. Review boundaries assigned to groups
2. Verify boundaries include target resources
3. Check boundary conditions

### Permission Evaluation Order

```
1. User authenticated? → No = Auth failure
2. User in any groups? → No = No permissions
3. Any policy grants permission? → No = Access denied
4. Any boundary restricts? → Yes = Check boundary scope
5. Access granted
```

### Common Permission Issues

| Issue | Symptom | Resolution |
|---|---|---|
| Not in group | No access at all | Add user to group via IdP |
| Wrong group | Partial access | Move to correct group |
| Policy too restrictive | Can't perform action | Modify or add policy |
| Missing environment | Can't access env | Add env to policy scope |

## 4. Policy Conflict Resolution

When multiple policies interact unexpectedly.

### How Policy Evaluation Works

1. Collect all policies from all user's groups
2. Evaluate each policy's conditions
3. **Any ALLOW** = access granted (unless explicitly denied)

4. **No matching ALLOW** = access denied

## Conflict Scenarios

| Scenario | Policy A | Policy B | Result |
|---|---|---|---|
| Both allow | ALLOW read | ALLOW write | ALLOW both |
| One allows | ALLOW read | No match | ALLOW read |
| Overlapping | ALLOW all envs | ALLOW prod only | ALLOW all envs |
| Condition mismatch | ALLOW (condition: A) | ALLOW (condition: B) | Depends on conditions |

## Debugging Policy Conflicts

### Step 1: List All Applicable Policies

1. Identify all groups user belongs to
2. List all policies assigned to those groups
3. Note overlapping permissions

### Step 2: Evaluate Each Policy

For each policy, check:
- Does it apply to the target resource?
- Does it grant the required permission?
- Are conditions met?

### Step 3: Identify Conflict

| Conflict Type | Indicator | Resolution |
|---|---|---|
| Unintended restriction | User in wrong group | Remove from group |
| Missing permission | No policy grants access | Add policy |
| Condition not met | Policy exists but doesn't apply | Adjust condition |
| Scope mismatch | Policy applies to wrong scope | Correct scope |

# 5. Boundary Troubleshooting

Boundaries restrict what data users can access within their permissions.

## Boundary Evaluation

```
User wants to access data
  ↓
Policy grants permission? → No = Access denied
  ↓ Yes
User has boundary assigned? → No = Access all data
  ↓ Yes
Data within boundary scope? → No = Data hidden
  ↓ Yes
Data returned
```

## Common Boundary Issues

| Issue | Symptom | Cause | Resolution |
|---|---|---|---|
| Too restrictive | See no data | Boundary too narrow | Expand boundary |
| Not restrictive enough | See too much | Boundary too broad | Narrow boundary |
| Segment applied | Some data missing | A segment the user selected hides data they are allowed to see | Clear the segment selector before debugging access |
| Environment excluded | No env access | Env not in boundary | Add environment |

## Boundary Debugging Steps

### Step 1: Identify Boundary

1. Find user's groups
2. Check boundaries assigned to groups
3. Note boundary scope (environment, storage, settings)

### Step 2: Verify Boundary Scope

For each domain:
- **Environment**: Which environments included?
- **Storage**: Which data types included?
- **Settings**: Which settings accessible?

**Step 3: Test Without Boundary**

Temporarily:
1. Add user to unbounded admin group
2. Test if access works
3. If yes, boundary is the issue
4. Remove from admin group

**Step 4: Adjust Boundary**

Modify boundary to include:
- Missing environments
- Missing security-context values
- Missing data types

# 6. SSO and Authentication Issues

When users can't login at all.

## SSO Troubleshooting

| Symptom | Check | Resolution |
|---|---|---|
| Redirect loop | IdP configuration | Verify ACS URL |
| SAML error | SAML response | Check attribute mapping |
| User not found | JIT provisioning | Enable auto-provisioning |
| MFA failure | MFA configuration | Verify MFA setup |
| Session expired | Session settings | Extend session duration |

## SAML Debugging

**Check SAML Response:**

1. Use browser dev tools → Network
2. Find SAML response during login
3. Decode and verify:
   - Subject/NameID (email)
   - Groups attribute
   - Conditions (timestamps)

**Common SAML Issues:**

| Issue | Indicator | Fix |
|---|---|---|
| Wrong NameID | Email doesn't match | Configure NameID format |
| Missing groups | Groups not in assertion | Add groups attribute in IdP |
| Expired assertion | Time mismatch | Sync clocks, extend validity |
| Invalid signature | Signature error | Update IdP certificate |

### Group Mapping Issues

If user logs in but has wrong permissions:

1. Check SAML groups in assertion
2. Verify group mapping configuration
3. Confirm IdP group names match exactly
4. Check for case sensitivity

## 7. Token and API Access Issues

When API tokens or OAuth clients don't work as expected.

### Token Issues

| Error | Cause | Resolution |
|---|---|---|
| `401 Unauthorized` | Token invalid/expired | Regenerate token |
| `403 Forbidden` | Missing scope | Add required scope |
| `404 Not Found` | Wrong endpoint or no data | Verify API path |

| Error | Cause | Resolution |
|---|---|---|
| Token rejected | Token revoked | Create new token |

## Token Debugging Steps

### Step 1: Verify Token Valid

1. Check token hasn't expired
2. Verify token not revoked
3. Confirm token is for correct environment

### Step 2: Check Scopes

1. List token's scopes
2. Compare to required scopes for API
3. Add missing scopes if needed

### Step 3: Test with curl

```
# Connectivity smoke test — pick an endpoint that survives the Gen3 upgrade
curl -H "Authorization: Api-Token &lt;TOKEN&gt;" \
  https://&lt;env&gt;.live.dynatrace.com/api/v2/settings/schemas
```

> **Do not use `/api/v2/entities` for this test.** It is flagged blocked at upgrade, so on a latest-Dynatrace tenant it fails for a reason that has nothing to do with the token — sending you to debug scopes and expiry when the endpoint itself is gone. `/api/v2/settings/schemas` carries forward and needs only `settings.read`, which makes it a cleaner probe. To check an entity actually resolves, use DQL (`smartscapeNodes`) rather than the classic Entities API — see FAQ entry 16.

## OAuth Client Issues

| Issue | Check | Resolution |
|---|---|---|
| Can't get token | Client credentials | Verify client ID/secret |
| Wrong permissions | Client scopes | Add required scopes |
| Token expired | Token lifetime | Request new token |

# 8. Diagnostic Queries

DQL queries to help diagnose IAM issues.

```
// Find recent access denied events
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "AUDIT_EVENT"
| filter event.outcome == "403"
| fields timestamp, user.id, resource, event.type, origin.address
| sort timestamp desc
| limit 25
```

```
// Recent authenticated API calls (credential type per call)
// Retitled 10/12/2026: authentication.type is set on every audited request,
so this lists API traffic.
// To list sign-ins only, filter event.type == "LOGIN".
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter isNotNull(authentication.type)
| fields timestamp, user.id, authentication.type, authentication.grant.type,
event.outcome
| sort timestamp desc
| limit 50
```

```
// Failed sign-in attempts (event.type LOGIN with outcome "failure")
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter event.type == "LOGIN" and event.outcome == "failure"
| fields timestamp, user.id, authentication.type, origin.address
| sort timestamp desc
| limit 50
```

```
// Recent IAM API activity in this environment (account-level policy or group
changes: account audit log)
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter in(event.type, {"POST", "PUT", "PATCH", "DELETE", "CREATE",
"UPDATE"}) and contains(resource, "iam")
| filter not startsWith(resource, "/lookups/")
| fields timestamp, user.id, event.type, resource
| sort timestamp desc
| limit 25
```

```
// Token-related events
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter authentication.type == "TOKEN" or contains(resource, "token")
| fields timestamp, user.id, event.type, resource, event.outcome
| sort timestamp desc
| limit 25
```

```
// All audit events in the last hour (immediate troubleshooting)
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-1h
| filter event.kind == "AUDIT_EVENT"
| fields timestamp, user.id, event.type, resource, event.outcome
| sort timestamp desc
| limit 50
```

## Troubleshooting Checklist

Use this checklist when debugging access issues:

```
AUTHENTICATION
☐ User can access IdP login page
☐ User can authenticate with IdP
☐ SAML assertion is valid
☐ User account exists in Dynatrace
☐ User account is active (not disabled)

GROUP MEMBERSHIP
☐ User is in expected groups
☐ Groups are synced from IdP (if using SCIM/JIT)
☐ No unexpected group memberships

POLICIES
☐ Groups have policies assigned
☐ Policies grant required permissions
☐ Policy conditions are met
☐ No conflicting policies

BOUNDARIES
☐ Boundary includes target environment
☐ Boundary includes target data type
☐ No overly restrictive boundaries

DATA
☐ Data exists in target timeframe
☐ No segment is selected that hides the data (segments filter views, they do
not restrict access)
☐ Query syntax is correct
```

## Escalation Path

When self-service troubleshooting doesn't resolve the issue:

| Level | Who | When |
|---|---|---|
| L1 | Team lead | Can't determine cause |
| L2 | IAM admin | Need policy/boundary changes |
| L3 | Platform team | Infrastructure/IdP issues |
| L4 | Dynatrace support | Product issues |

### Information for Escalation

Include in escalation:

```
☐ User email/identifier
☐ Exact error message
☐ Steps to reproduce
☐ Timestamp of issue
☐ What was already checked
☐ Audit log excerpts
☐ Screenshots
```

## Series Conclusion

Congratulations! You've completed the IAM series.

### Series Recap

| Notebook | Key Topics |
|---|---|
| **01: Governance Foundations** | Framework, roles, models |
| **02: SSO and Authentication** | SAML, IdP integration, MFA |
| **03: Group Architecture** | Hierarchy, naming, SAML mapping |
| **04: Policy Authoring** | Syntax, conditions, GitOps |
| **05: Boundary Design** | Three domains, isolation patterns |
| **06: User Lifecycle** | SCIM, JIT, tokens |
| **07: Audit and Compliance** | Audit queries, SOC2/SOX/HIPAA |
| **08: Multi-Environment** | Account vs env, break-glass |
| **09: Troubleshooting** | Methodology, common issues |
| **10: Templated Policy-Group Assignments** | Parameterized policies, IAM API bindings, Monaco |
| **11: Policy Persona Workshop** | Persona-based policy design, schema audit, domain mapping |

| Notebook | Key Topics |
|---|---|
| **12: API Provisioning & Validation** | Provisioning scripts, binding reports, cleanup, validation DQL |
| **95: Terraform IAM Provisioning (LAB)** | Groups, policies, boundaries and bindings as Terraform |
| **96: Python IAM Provisioning (LAB)** | Step-by-step Account Management API provisioning in Python |
| **99: Best Practice Summary** | 72 best practices with priorities and settings |

### Ongoing Maintenance

| Task | Frequency |
|---|---|
| Access reviews | Quarterly |
| Policy review | Semi-annually |
| Boundary review | Semi-annually |
| Token rotation | Per policy (90 days typical) |
| Audit log review | Monthly |
| Documentation update | As needed |

---

## Summary

In this notebook, you learned:

- Systematic troubleshooting methodology
- Common access issue categories
- Permission denied debugging steps
- Policy conflict identification and resolution
- Boundary troubleshooting techniques
- SSO and authentication debugging
- Token and API access troubleshooting

- Diagnostic DQL queries

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## References

- IAM Troubleshooting
- SAML Configuration
- API Authentication

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